Microsoft Elevate

CAPSTONE PROJECT

Amazon Sales Analysis using Power BI

PRESENTED BY

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OUTLINE:

- Problem Statement
- Proposed System/Solution
- System Development Approach
- Algorithm & Deployment
- Result (Dashboard Output)
- Conclusion
- Future Scope
- References

PROBLEM STATEMENT:

E-commerce platforms like Amazon generate a large volume of sales data every day. Analyzing this data manually becomes difficult and time-consuming for businesses. Without proper data analysis, it is challenging to identify sales trends, product performance, and regional sales patterns. Therefore, there is a need for an effective data analysis system that can transform raw sales data into clear and meaningful insights to support better business decision-making.

PROPOSED SOLUTION:

- **Proposed Solution**
- The proposed system aims to analyze Amazon sales data and convert it into meaningful business insights using **Power BI**.
- The solution focuses on transforming raw sales data into interactive dashboards that help users understand sales performance, product trends, and regional distribution.
- **Data Collection**
- • Collect Amazon sales dataset containing information such as product category, sales amount, profit, region, and order date.
  • Import the dataset into Power BI for analysis.
- **Data Preprocessing**
- • Clean the dataset by removing missing or inconsistent values.
  • Transform and format columns using **Power Query Editor**.
  • Create calculated columns and measures using **DAX**.
- **Dashboard Development**
- • Design interactive dashboards to visualize sales insights.
  • Use charts such as **bar charts, line charts, pie charts, and KPI cards**.
  • Add filters and slicers to allow users to explore data dynamically.
  The developed dashboard enables businesses to easily monitor sales trends, analyze product performance, and make better data-driven decisions.

SYSTEM APPROACH:

- Tools & Technologies Used:
- - Microsoft Power BI
- - Excel Dataset
- - Power Query
- - DAX (Data Analysis Expressions)

- Development Steps:
- 1. Data Collection
- 2. Data Cleaning
- 3. Data Transformation
- 4. Data Modeling

ALGORITHM & DEPLOYMENT:

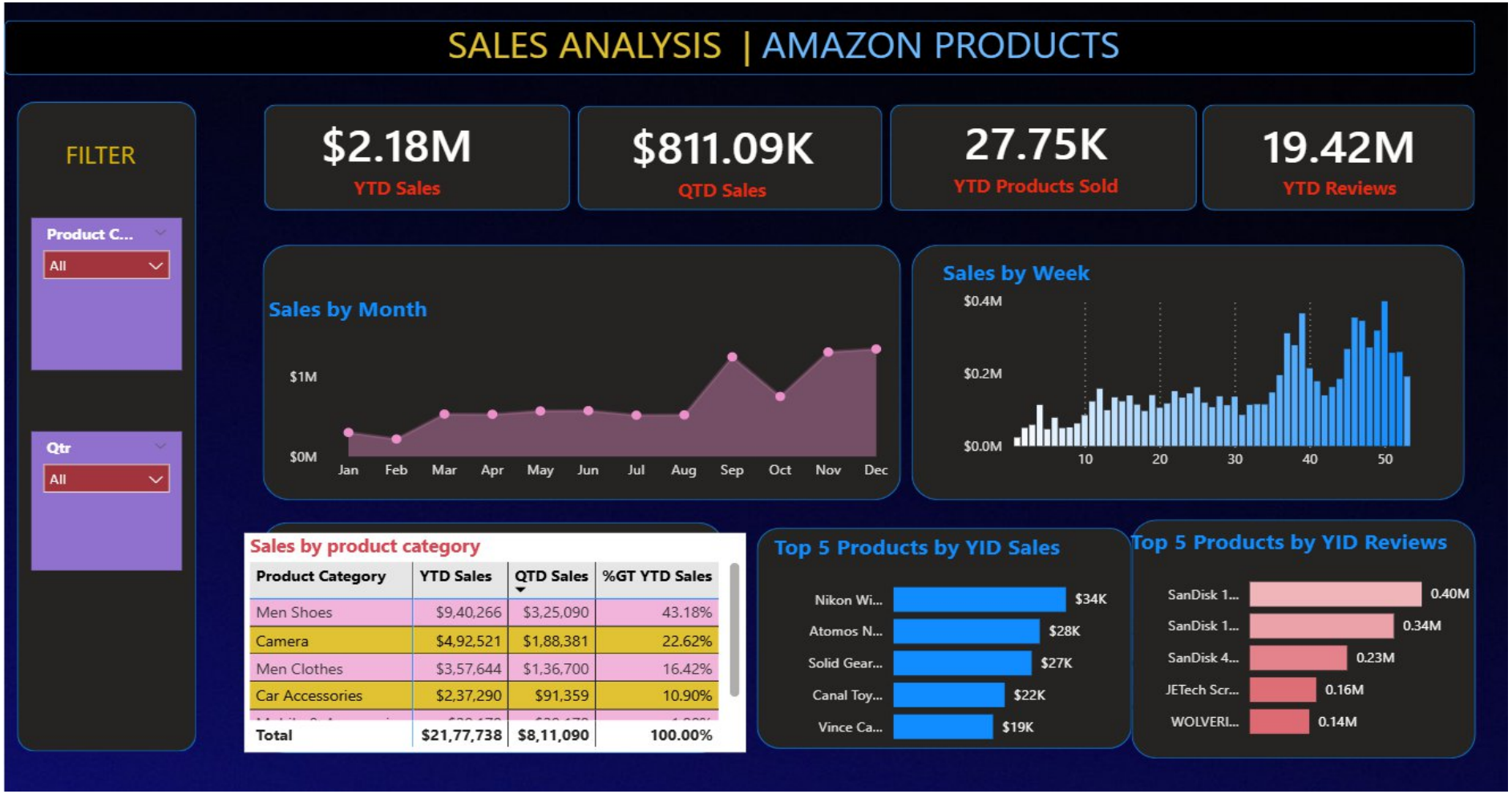
Data Analysis Process

- Sales dataset imported into **Power BI**
- Data cleaned and transformed using **Power Query**
- Created calculated measures using **DAX**
- Data relationships established for analysis
- Visualizations created to analyze sales trends and performance

Deployment

- The dashboard is developed using **Power BI Desktop**
- Interactive reports allow users to filter and explore sales data
- The dashboard can be published on **Power BI Service** for online access and sharing

RESULT:



Key Insights:

- Total Sales
- Category Performance
- Monthly Trend
- Regional Sales

CONCLUSION:

The **Amazon Sales Analysis dashboard** developed using Microsoft Power BI successfully transforms raw sales data into meaningful and interactive visual insights. The project helps analyze important factors such as sales performance, product category trends, and regional distribution. By using data visualization techniques, the dashboard makes it easier for users to understand large datasets and identify key business patterns. Overall, this project demonstrates how business intelligence tools can support better data-driven decision making and improve business analysis efficiency.

FUTURE SCOPE:

In the future, the Amazon Sales Analysis dashboard can be enhanced by integrating **real-time sales data** to provide live business insights. Advanced analytics techniques and predictive models can also be applied to forecast future sales trends and demand. The dashboard can be expanded by including **customer behavior analysis and product recommendation insights**. Additionally, the system can be deployed on cloud platforms such as **Microsoft Power BI Service** to enable online access, sharing, and collaboration for business users.

REFERENCES:

The development of this project was supported by various learning resources and documentation related to business intelligence and data visualization. Official documentation and tutorials from **Microsoft Power BI** and **Microsoft Learn** were used to understand data modeling, Power Query, and DAX functions. The dataset used for analysis was obtained from **Kaggle**, which provides publicly available datasets for data analysis projects. Additional online resources and articles on business intelligence and data visualization were also referred to during the development of this Amazon Sales Analysis dashboard.

GitHub Link: https://github.com/AmanChouhan074/Amazon-sales-analytics-powerbi-dashboard

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Thank You